8	A body of mass 90 kg travels along a horizontal road at a speed of 5.0 m s^{-1}. It then accelerates at 1.5 m s^{-2}. At the time it begins to accelerate, the total resistive force acting on the car is 200 N. What total output power is developed by the car as it begins the acceleration?	
	A	680 W
	B	1700 W
	C	4400 W
	D	5400 W

Answer: B

ma = F – resistive force

F = 90 x 1.5 + 200 = 335 N

Therefore power required to give the 90 kg, 5.0 m s^{-1} body to accelerate at 1.5 m s^{-2} is

F x v

= 335 x 5

= 1675 W

= 1700 W